



Certificate of Calibration

Ocean Sonics, Ltd.

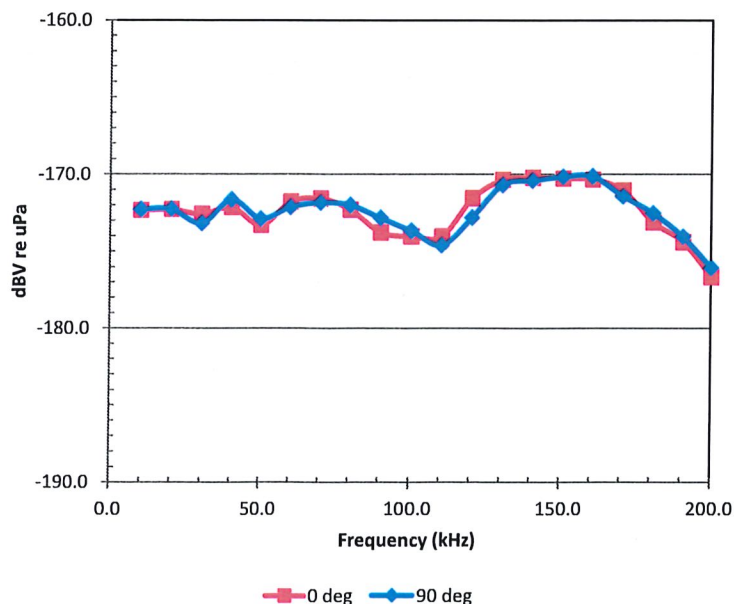
Calibration Certificate Number: C11041

Test Result: 10 kHz to 100 kHz: -172.8 ± 1.2
10 kHz to 200 kHz: -173.4 ± 3.3

Model Number SB35-ETH
Serial Number 1249
Manufacture Date 20-Mar-2013
Measurement Date 26-May-2026
Certificate Date 26-May-2026
Sensitivity @ 26 Hz -170.1 dB re V / uPa
Case Type Titanium
Element Manufacturer Geospectrum
Element Model M24HF
Element Serial I001092
Preamp Model 04-300435-09
Calibrated By D.Lee
Work Order Number W0-90
Test Type RX Sensitivity
Test Procedure Complex RMS
Test Location Tank #3, 1 m
Water Temperature 16 °C

Projector Manufacturer Ocean Sonics
Projector Model TH2-SER-4F
Projector Serial 2225
Measurement Distance 1 m
Output Level 129.7 dB re uPa @ 1 m
Tone Burst 100.0 us / 300 ms
Reference Manufacturer Ocean Sonics
Reference Model RB9-ETH
Reference Serial 6465
Primary Calibration 2023-07-04
Preamp Manufacturer Ocean Sonics
Preamp Model 04-300451-01
Preamp Serial Z88
Preamp Gain 36 dB
ADC Manufacturer Ocean Sonics
ADC Model Number 04-300427-01
ADC Serial Number Z87

Receive Sensitivity



Frequency kHz	Sensitivity [dBV re uPa]	
	0 deg	90 deg
10.0	-172.3	-172.3
20.1	-172.3	-172.3
30.1	-172.6	-173.2
40.2	-172.2	-171.7
50.2	-173.3	-172.9
60.2	-171.8	-172.2
70.3	-171.6	-171.9
80.3	-172.3	-172.0
90.4	-173.8	-172.9
100.4	-174.1	-173.7
110.4	-174.0	-174.7
120.5	-171.6	-172.9
130.5	-170.3	-170.7
140.5	-170.2	-170.4
150.6	-170.3	-170.2
160.6	-170.3	-170.2
170.7	-171.1	-171.5
180.7	-173.1	-172.6
190.7	-174.4	-174.1
200.0	-176.7	-176.1



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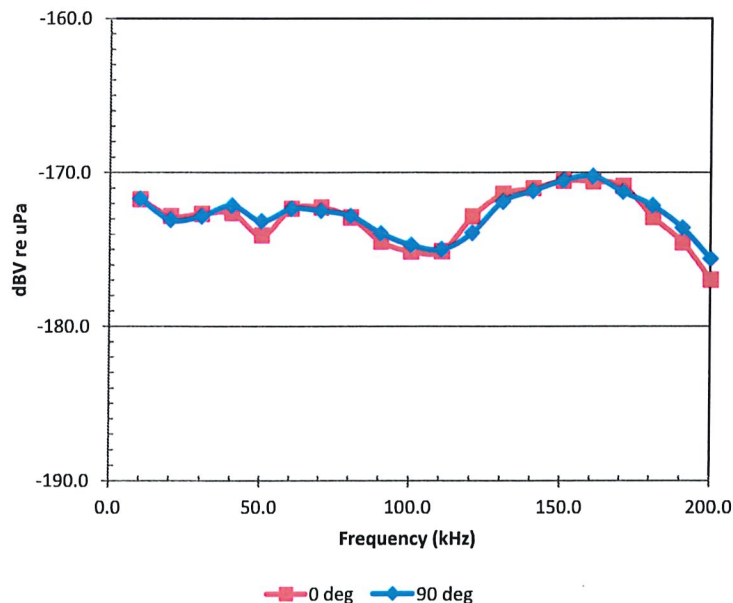
Ocean Sonics, Ltd.

Calibration Certificate Number: **C10909**

Test Result: 10 kHz to 100 kHz: -173.4 ± 1.7
10 kHz to 200 kHz: -173.6 ± 3.4

Model Number	SB35-ETH	Projector Manufacturer	Ocean Sonics
Serial Number	1249	Projector Model	TH2-SER-4F
Manufacture Date	20-Mar-2013	Projector Serial	2225
Measurement Date	27-Apr-2026	Measurement Distance	1 m
Certificate Date	27-Apr-2026	Output Level	130 dB re uPa @ 1 m
Sensitivity @ 26 Hz	-170.1 dB re V / uPa	Tone Burst	100.0 us / 300 ms
Case Type	Titanium	Reference Manufacturer	Ocean Sonics
Element Manufacturer	GeoSpectrum	Reference Model	RB9-ETH
Element Model	M24HF	Reference Serial	6465
Element Serial	I001092	Primary Calibration	2023-07-04
Preamplifier Model	04-300435-09	Preamplifier Manufacturer	Ocean Sonics
Calibrated By	D.Lee	Preamplifier Model	04-300451-01
Work Order Number	W0-90	Preamplifier Serial	Z88
Test Type	RX Sensitivity	Preamplifier Gain	36 dB
Test Procedure	Complex RMS	ADC Manufacturer	Ocean Sonics
Test Location	Tank #3, 1 m	ADC Model Number	04-300427-01
Water Temperature	16 °C	ADC Serial Number	Z87

Receive Sensitivity



Frequency kHz	Sensitivity [dBV re uPa]	
	0 deg	90 deg
10.0	-171.7	-171.7
20.1	-172.8	-173.1
30.1	-172.7	-172.9
40.2	-172.6	-172.2
50.2	-174.1	-173.2
60.2	-172.4	-172.4
70.3	-172.3	-172.5
80.3	-172.9	-172.8
90.4	-174.5	-174.0
100.4	-175.1	-174.7
110.4	-175.1	-175.0
120.5	-172.8	-173.9
130.5	-171.4	-171.9
140.5	-171.0	-171.2
150.6	-170.5	-170.5
160.6	-170.6	-170.3
170.7	-170.9	-171.3
180.7	-172.9	-172.2
190.7	-174.5	-173.6
200.0	-177.0	-175.6